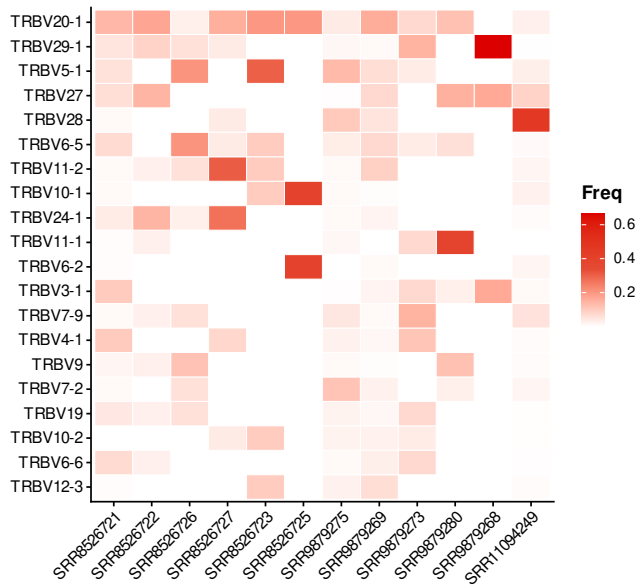
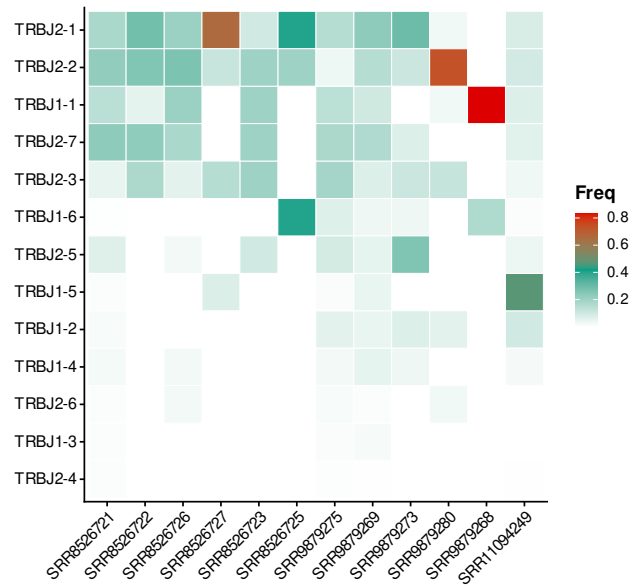
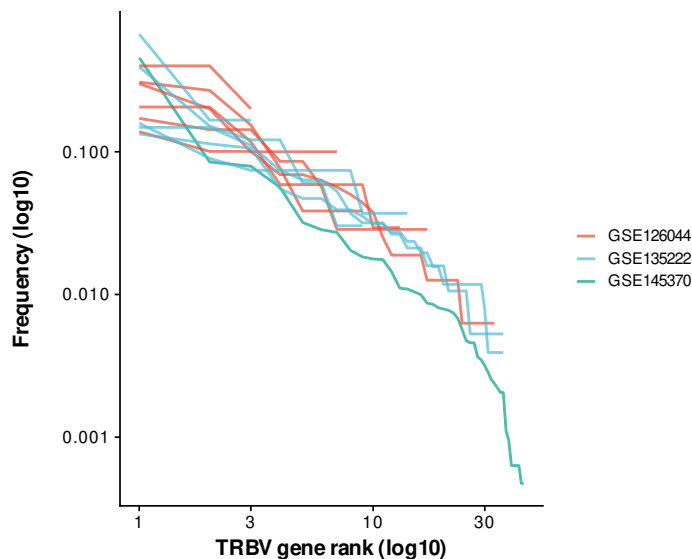


A**A. Top-20 TRBV gene usage****B****B. TRBJ gene usage (all)****C****C. V-gene rank–frequency****D****D. TRBV → TRBJ pairing (top 12 V × top 6 J)**